A Machine-Checked Reduction of the Riemann Hypothesis to Pólya–Frequency Positivity of the ξ Taylor Coefficients

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Revised manuscript — October 10, 2026
Local revision 1.0.1-draft of version 1.0.0 (October 7, 2026)
Reserved manuscript DOI: 10.5281/zenodo.20574510

Abstract

The classical Laguerre–Pólya approach expresses the Riemann Hypothesis (RH) as an all-orders positivity problem for the Taylor coefficients of Riemann's ξ function. We present a Lean 4/Mathlib formalization of conditional reductions within this approach. Given explicitly named external inputs for the Pólya–Jensen criterion, the Aissen–Schoenberg–Whitney/Edrei characterization, limit closure, and the converse, the central theorem identifies RH with total nonnegativity of the sign-normalized coefficient Toeplitz matrix. This equivalence is classical; the contribution is a checked dependency structure, together with algebraic coefficient and moment identities. We distinguish ordinary coefficients μ_n from the classical Jensen coefficients $n!\mu_n$, explain why Mathlib's entire pole-subtracted completed zeta is not itself the correct zero-set target, and record a separate pseudo-Hermitian eigenvector reduction. Exact rational coefficient enclosures and FLINT/Arb ball arithmetic certify positivity of 1816 contiguous Toeplitz minors of orders at most 16, using coefficients through index 120, and strict log-concavity through index 118. These computations calibrate the conventions within ranges already covered by classical positivity results; they do not establish uniformity in the order. A theorem ledger and replay instructions specify the formal and numerical trust boundaries. **No proof of RH is claimed.**

Keywords: Riemann Hypothesis; Riemann xi function; Pólya-frequency sequences; total positivity; Jensen polynomials; Lean formalization; rigorous ball arithmetic.

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1 Introduction and scope

Disclaimer. This paper is a *formalization and reduction study*, not a proof of the Riemann Hypothesis, and it makes no claim to be one. Every implication labelled "verified" is checked mechanically; every classical theorem we rely upon but do not formalize is cited explicitly as an *external input* and is never silently assumed. The defensible contributions are:

1. a correct, machine-checked and *tight* formal version of the classical reduction of RH to a single sign-correct positivity statement;
2. the identification and repair of a target-function error that arises naturally when working with Mathlib's completed zeta;
3. an independent operator-positivity reduction;
4. reproducible, ball-arithmetic consistency checks of the reduction's premise in a finite range.

We do not claim that the equivalence, the Turán inequalities or finite-order Pólya-frequency positivity of the ξ coefficients are new; see Section 8.

Version history and relation to later work. This manuscript was drafted in June 2026 (reserved DOI `10.5281/zenodo.20574510`). The version 1.0.0 source records four changes:

1. it replaces two-precision floating cross-checks with ball-arithmetic certification (Section 5);
2. it removes earlier claims that the computation exceeded what is known; Katkova's theorem already yields nonnegativity of every minor of order ≤ 43;
3. it records that frontier item C of the June draft is settled in the literature;
4. it ties every cited Lean theorem to an archived commit.

The subsequent development is reported separately. A formalized Fekete-type closure from strict contiguous minors to full Pólya-frequency positivity, together with effective and Jensen receptor interfaces, appears in [25]. An independent prime-defined Weil-form programme appears in [26]. The present revision adds the coefficient-normalization dictionary, explicit bridge statements, a theorem ledger and reproducibility appendix. It also corrects the distinction between contiguous minors and arbitrary minors, and between moment ratios and coefficient ratios. A reserved DOI is a manuscript identifier; this revision does not assert that a new version has been deposited.

Organization. Section 2 fixes the conventions. Sections 3 and 4 give the target correction and conditional reductions. Sections 5 and 6 describe the finite certificates and remaining analytic obligations. Section 7 and Appendix A identify the artifacts and verification procedure.

2 Preliminaries and normalization

The object. Riemann's entire ξ function

$$\xi(s) \;=\; \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\, \Gamma(s/2)\, \zeta(s) \tag{1}$$

is entire, and its zeros are exactly the nontrivial zeros of ζ; RH asserts they all lie on $\operatorname{Re} s = \frac{1}{2}$. On the critical line, the real even function $\Xi(t) = \xi(\frac{1}{2} + it)$ has a Taylor expansion

$$\Xi(t) \;=\; \sum_{n\geq 0} b_n\, t^{2n}, \qquad b_n \in \mathbb{R}, \tag{2}$$

and we set the *sign-normalized sequence* $\mu_n := (-1)^n b_n$.

Two classical pillars.

- *Pólya–Jensen* [3, 7]: RH is equivalent to hyperbolicity (all roots real) of every Jensen polynomial $J^{\mathrm{cl}}_{d,n}(X) = \sum_{k=0}^{d} \binom{d}{k} \gamma_{n+k} X^k$, where $\gamma_n = n!\mu_n$ is the exponential-series coefficient of the folded function defined below [15].
- *Aissen–Schoenberg–Whitney / Edrei* [4, 5, 6]: a real sequence is a *Pólya-frequency* (PF) sequence—all minors of the Toeplitz matrix $[a_{i-j}]$ are nonnegative—iff its generating function is $Cz^{\ell}e^{az}\prod(1+\alpha_i z)/\prod(1-\beta_j z)$ with $C, a, \alpha_i, \beta_j \geq 0$, $\ell \geq 0$, and $\sum_i \alpha_i + \sum_j \beta_j < \infty$. For an entire generating function the denominator is absent: this is the nonnegative-coefficient, type-I subclass of the Laguerre–Pólya class, not the unrestricted class.

Together: *RH* $\iff$ (μ_n) *is a Pólya-frequency sequence.* This consequence is classical; Katkova [13] states it explicitly for $\xi_1(z) = \xi(\frac{1}{2} + \sqrt{z}) = \sum_n \mu_n z^n$. She also proves that (μ_n) is a Pólya-frequency sequence of every finite order up to an explicit bound, using a zero-free sector theorem of Schoenberg. Her printed proof gives order 43, while her statement gives 44; we use 43. Here μ_n coincides with Katkova's coefficients and with $\lambda_n = \xi^{(2n)}(\frac{1}{2})/(2n)!$ of [20].

The normalization dictionary in Table 1 separates three coefficient sequences which cannot be interchanged in the Jensen criterion. Define the folded entire function by its power series, so no choice of a square-root branch is involved:

$$F(z) = \sum_{n\geq 0} \mu_n z^n = \sum_{n\geq 0} \gamma_n \frac{z^n}{n!}, \qquad \xi(\tfrac{1}{2} + w) = F(w^2), \qquad \Xi(t) = F(-t^2). \tag{3}$$

Its order is 1/2. Under RH its zeros are negative, and its genus-zero product lies in the type-I class above. Conversely, negative real zeros of F give real zeros of Ξ, hence the critical-line zero condition for ξ. The analytic identification and classical characterizations are background theorems; the formal bridges below carry the corresponding hypotheses.

Table 1: Coefficient and index conventions used throughout. Negative indices are zero extensions, not truncated subtraction in $\mathbb{N}$.

Symbol	Definition	Lean identifier or role
b_n	$\Xi^{(2n)}(0)/(2n)!$	`XiCoeff`
μ_n	$(-1)^n b_n$	`XiMomentCoeff`
γ_n	$n!\mu_n$	`XiJensenCoeff`
$J^{\mathrm{cl}}_{d,n}$	$\sum_{j=0}^{d} \binom{d}{j}\gamma_{n+j}X^j$	`ClassicalJensenPoly`
$Q_{d,n}$	$\sum_{j=0}^{d} \binom{d}{j}b_{n+j}X^j$	legacy `JensenPoly`
$D_k(m)$	$\det[\mu_{m+i-j}]_{0\le i,j<k}$	`XiToeplitzContigMinor`
μ_j, $j<0$	0	zero-extended Toeplitz entries

In particular, the diagnostic polynomial $Q_{d,n}$ is not the classical Jensen polynomial of F. The factorial in γ_n changes the hyperbolicity inequalities. Assertions below about Q are identified separately and are not substituted into `PolyaJensenBridge`.

Pólya's positive kernel. By Pólya's representation [2], $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(ut)\,du$, in the convention

$$\Phi(u) = \sum_{j\ge 1} \left(4\pi^2 j^4 e^{9u/2} - 6\pi j^2 e^{5u/2}\right) e^{-\pi j^2 e^{2u}}, \qquad u \ge 0. \tag{4}$$

Each summand is positive because $2\pi j^2 e^{2u} > 3$. The double-exponential decay permits differentiation under the integral at every fixed order, so

$$b_n = (-1)^n \frac{2M_n}{(2n)!}, \qquad M_n := \int_0^\infty \Phi(u)\, u^{2n}\, du > 0, \qquad \mu_n = \frac{2M_n}{(2n)!} > 0. \tag{5}$$

The corresponding generating-function identity is $F(z) = 2\int_0^\infty \Phi(u)\cosh(u\sqrt{z})\,du$, with the integrand understood through its entire power series. The factor 2 agrees with the coefficient generator and is essential to the moment normalization.

3 The target-function correction: ξ, not Λ_0

We record an error and its repair, because catching it is part of the contribution. Mathlib's *entire* completed zeta $\Lambda_0 = \texttt{completedRiemannZeta}_0$ satisfies $\Lambda = \texttt{completedRiemannZeta} = \Lambda_0 - 1/s - 1/(1-s)$.

Proposition 1 (Wrong target)**.** *The zeros of Λ_0 are not the nontrivial zeros of ζ; in fact Λ_0 has no zeros on the critical line at all. At a nontrivial zero ρ one has $\Lambda(\rho) = 0$ but $\Lambda_0(\rho) = \frac{1}{\rho} + \frac{1}{1-\rho} = \frac{1}{\rho(1-\rho)} \neq 0$. On the critical line,*

$$\Lambda_0(\tfrac{1}{2} + it) = \frac{1 - 2\,\Xi(t)}{t^2 + \frac{1}{4}} \;\ge\; \frac{1 - 2\,\xi(\frac{1}{2})}{t^2 + \frac{1}{4}} > 0 \qquad (t \in \mathbb{R}), \tag{6}$$

with $\xi(\frac{1}{2}) = 0.4971207781\ldots < \frac{1}{2}$.

Proof. Since $\Lambda = \Lambda_0 - 1/(s(1-s))$ and $\xi(s) = \frac{1}{2}s(s-1)\Lambda(s)$, one has $\Lambda_0(s) = (1-2\xi(s))/(s(1-s))$, and $s(1-s) = t^2 + \frac{1}{4}$ at $s = \frac{1}{2} + it$. Pólya's kernel is positive [2], so $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(tu)\,du$ obeys $|\Xi(t)| \le 2\int_0^\infty \Phi = \Xi(0) = \xi(\frac{1}{2})$. The certified coefficient $\mu_0 = \xi(\frac{1}{2})$ (Section 5) lies in a ball around 0.4971207781, strictly below $\frac{1}{2}$. □

Thus the level set $\{\Xi = \frac{1}{2}\}$ named by the naive target is empty on the real line, and every zero of Λ_0 lies off the critical line. Figure 1 shows the gap. It is small: the minimum of $1 - 2\Xi(t)$ is 0.00576, at $t = 0$.

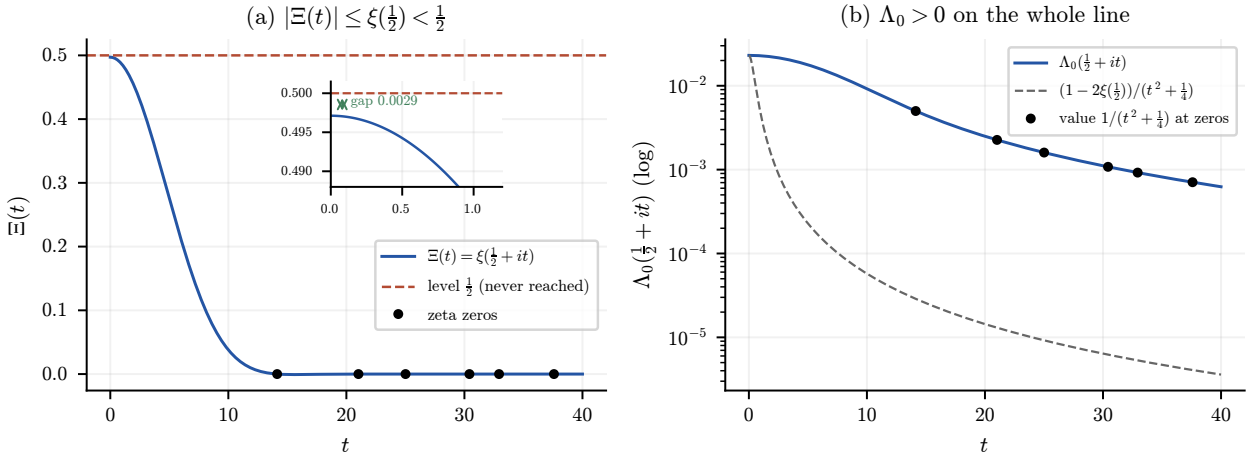


Figure 1: The Mathlib target trap. (a) $\Xi(t) = \xi(\frac{1}{2} + it)$ crosses zero at the zeta zeros (dots) but never reaches the level $\frac{1}{2}$; the inset magnifies the gap $\frac{1}{2} - \xi(\frac{1}{2}) \approx 0.0029$ at $t = 0$. (b) $\Lambda_0(\frac{1}{2} + it)$ on a logarithmic scale, with the lower bound of (6) dashed. At each zeta zero $\Lambda_0 = 1/(t^2 + \frac{1}{4})$, which is nonzero. Curves are mpmath evaluations at 30 digits.

The remedy is to multiply Λ by $s(s-1)/2$ away from $s = 0, 1$, where the factor is nonzero, and extend analytically across those two points. The Lean definition avoids multiplying pointwise pole values by zero:

$$\texttt{xiCompleted}(s) = \tfrac{1}{2} + \tfrac{1}{2}s(s-1)\Lambda_0(s). \tag{7}$$

This entire function takes value 1/2 at 0 and 1, and equals the usual product elsewhere (`xiCompleted_eq_product`). We verify that the corrected slice has the correct zero set:

Theorem 2 (`Xi_eq_zero_iff_riemannZeta`; verified)**.** *For all* $t \in \mathbb{R}$, $\Xi(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.

Proof (mechanized). We define $\Xi(t) = -\frac{t^2+1/4}{2} Z(t)$ where $Z(t) = \mathrm{Re}\,\Lambda(\frac{1}{2} + it)$ is the verified real critical-line slice. The scalar $-\frac{t^2+1/4}{2}$ is a nonzero real, so $\Xi(t) = 0 \iff Z(t) = 0 \iff \Lambda(\frac{1}{2} + it) = 0 \iff \zeta(\frac{1}{2} + it) = 0$ (the last step using $\Lambda = \Gamma_{\mathbb{R}} \cdot \zeta$ with $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \neq 0$). All steps are checked in Lean. □

All total-positivity, Jensen, and moment results below are stated for this corrected ξ.

4 Verified reductions

Throughout, "verified" means the Lean theorem reduces to {propext, Classical.choice, Quot.sound}, confirmed by `#print axioms`. Definitions of type `Prop` that encode classical theorems we do not formalize are *named external inputs*; they are never asserted, only taken as explicit hypotheses.

Figure 2 shows how the verified pieces fit together. The classical inputs close a cycle through the full PF condition, so that condition is pinned to RH from both sides; everything else in this section is a checked consequence or an independent conditional route.

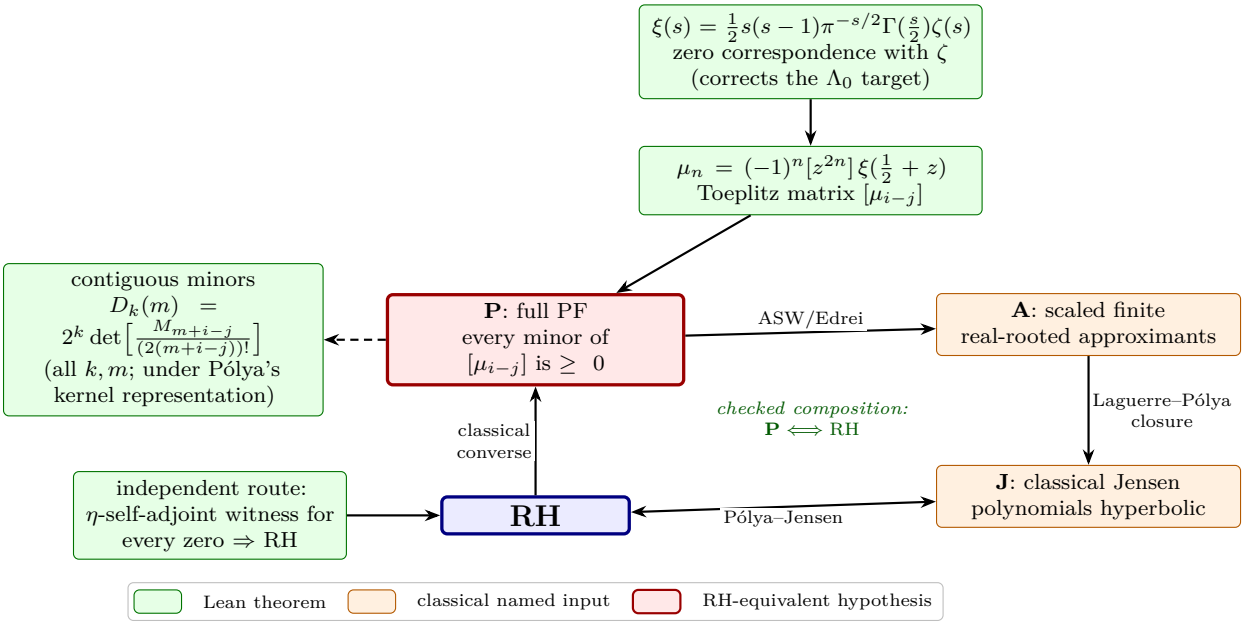


Figure 2: Logical shape of the verified reduction. Green boxes are Lean theorems depending only on Lean's three foundational axioms; amber boxes are named classical inputs, passed as explicit hypotheses and never asserted; the rose box is the one non-classical hypothesis, and Theorem 5 proves it equivalent to RH. The moment ladder is Theorem 7; the operator route is Theorem 10.

4.1 The Pólya-frequency / Toeplitz route

Definition 3 (`XiToeplitzTotalPositive`)**.** The sign-normalized sequence (μ_n) is *Pólya-frequency total positive* if every minor of the Toeplitz matrix $[\mu_{i-j}]$ (over all strictly monotone row and column index selections) is nonnegative.

Here "total positive" follows the historical PF terminology used in the code; mathematically the condition is *total nonnegativity*. Explicitly, for all $k \in \mathbb{N}$ and strictly increasing $r, c\colon \{0, \dots, k-1\} \to \mathbb{N}$,

$$\det[\mu_{r_i - c_j}]_{i,j<k} \geq 0, \qquad \mu_\ell = 0 \text{ for } \ell < 0. \tag{8}$$

The contiguous determinants $D_k(m)$ select $r_i = m + i$, $c_j = j$. Thus full PF implies all contiguous inequalities. A converse from merely nonnegative contiguous determinants is not proved by the extraction lemmas in this paper. The later strict-minor closure theorem cited above has additional hypotheses and must be invoked explicitly when used.

Proposition 4 (Low-order rungs; verified)**.** *Under `XiToeplitzTotalPositive`: (i) every* 1×1 *minor gives* $\mu_n \geq 0$*; (ii) the contiguous* 2×2 *minor equals the Turán determinant,* $\mu_{n+1}^2 - \mu_n \mu_{n+2}$*, so its nonnegativity is exactly the Turán inequality* $b_n b_{n+2} \leq b_{n+1}^2$*; (iii) for every order* k *and offset* m *the contiguous minor* $\det[\mu_{m+i-j}]_{i,j<k} \geq 0$.

We further prove (Lemma `xiToeplitzMinor3_sign_eq`) that the order-3 signed minor equals $(-1)^n$ times the unsigned b-determinant, exhibiting the correct alternating-positivity structure; and

we document that the Hankel/moment route used in an earlier draft is the *wrong* machine, since Hankel positivity forces *log-convexity*, the opposite of the (true) Turán log-concavity.

4.2 Tightness of the reduction

Let the three classical external inputs be: `XiToeplitzTPToScaledFiniteBridge` (Edrei–ASW), `XiLaguerrePolyaClosureBridge` (LP closure), and `PolyaJensenBridge`; and let `XiToeplitzTPFromRH` be the classical converse (RH $\Rightarrow$ ξ is LP $\Rightarrow$ its coefficients are PF).

Theorem 5 (`xiToeplitzTotalPositive_iff_riemannHypothesis`; verified)**.** *Given the three classical bridges and the classical converse,*

$$\mathit{XiToeplitzTotalPositive} \iff \mathrm{RH}.$$

Proof (mechanized). The forward direction composes the ASW/Edrei and closure inputs with the reverse implication in `PolyaJensenBridge`. The reverse direction is `XiToeplitzTPFromRH`. All four are arguments of the theorem; none is supplied by its proof. Writing P for full PF, A for the scaled finite-approximation target, and J for classical Jensen hyperbolicity, the checked composition is

$$(P \Rightarrow A),\ (A \Rightarrow J),\ (\mathrm{RH} \Leftrightarrow J),\ (\mathrm{RH} \Rightarrow P) \quad\Longrightarrow\quad (P \Leftrightarrow \mathrm{RH}).$$

The finite-approximation witness requires locally uniform convergence on $\mathbb{C}$, in addition to convergence on $\mathbb{R}$; pointwise real convergence alone does not meet this interface. □

Remark 6 (The honesty linchpin)*.* Theorem 5 is the classical equivalence [3, 4, 5, 13] in machine-checked form. Its value is not mathematical novelty. It mechanically pins down the one hypothesis that is not classical, and shows that this input is *equivalent* to RH. Hence declaring it an axiom is logically identical to `axiom rh : RiemannHypothesis`: it does not constitute a proof, and `#print axioms` would expose a newly declared axiom. An equivalent criterion can guide a proof, but a proof still has to establish that criterion without assuming RH. The axiom footprint checks dependencies of the conditional theorem; it does not discharge its explicit hypotheses.

4.3 The complete moment-determinant ladder

Under Pólya's representation (the named external input `XiMomentKernelRep`, encoding (5)), the PF condition transfers to explicit moment determinants at *every* order.

Theorem 7 (`xiToeplitzContigMinor_eq_of_kernelRep`; verified)**.** *For all $k, m \in \mathbb{N}$,*

$$\det\big[\mu_{m+i-j}\big]_{i,j<k} \;=\; 2^k \cdot \det\left[\frac{M_{m+i-j}}{(2(m+i-j))!}\right]_{i,j<k}.$$

Every entry with negative subscript on the right is defined to be zero. Consequently (`xiToeplitzContigMinor_nonneg_iff_moment_of_kernelRep`), for each k, m, nonnegativity of this contiguous coefficient determinant is equivalent to nonnegativity of the corresponding weighted moment determinant.

Proof (mechanized). Each entry of the signed Toeplitz matrix equals 2 times the corresponding factorial-weighted moment entry (by (5)); the determinant is then 2^k times the moment determinant by multilinearity (`Matrix.det_smul`, $\#\,\mathrm{Fin}\,k = k$). Positivity of 2^k gives the equivalence. □

The displayed Lean theorem concerns contiguous blocks. The same entrywise scaling gives an ordinary mathematical identity for arbitrary increasing row and column selections, with the same factor 2^k. Only when *all* such selections are quantified does this become a full PF reformulation. Positivity of the kernel gives $M_n > 0$ and positive Hankel moment matrices; it does not by itself prove these weighted Toeplitz inequalities.

The order-2 mechanism is quantified by the *slack factor* $s_n = \dfrac{(2n+3)(2n+4)}{(2n+1)(2n+2)}$, for which we verify $s_n > 1$ (`one_lt_slackFactor`) and $s_{n+1} < s_n$ (`slackFactor_succ_lt`); thus $s_n \downarrow 1$. To prevent confusing coefficient ratios with moment ratios, put

$$H_n = \frac{M_{n+1}^2}{M_n M_{n+2}}, \qquad R_n = \frac{\mu_{n+1}^2}{\mu_n \mu_{n+2}} = s_n H_n. \tag{9}$$

Moment log-convexity gives $H_n \le 1$, while coefficient log-concavity is $R_n \ge 1$, equivalently $H_n \ge 1/s_n$. These are compatible statements. The elementary slack gap is

$$1 - \frac{1}{s_n} = \frac{8n+10}{(2n+3)(2n+4)} \sim \frac{2}{n}. \tag{10}$$

The threshold $1/s_n$ applies to H_n, not to R_n.

4.4 Order two: a diagnostic quadratic and the classical normalization

Theorem 8 (`jensenPoly_two_hyperbolic_of_turan`; verified)**.** *If $b_{n+2} \neq 0$ and the Turán inequality $b_n b_{n+2} \le b_{n+1}^2$ holds, then the degree-2 diagnostic polynomial $Q_{2,n}$ is hyperbolic (all complex roots real). Hence (`allJensenTwo_hyperbolic_of_turan2All`) if all coefficients are nonzero and all Turán inequalities hold, every $Q_{2,n}$ is hyperbolic.*

Proof (mechanized). $Q_{2,n}$ maps to $b_{n+2}z^2 + 2b_{n+1}z + b_n$ over $\mathbb{C}$. Completing the square, $(b_{n+2}z + b_{n+1})^2 = b_{n+1}^2 - b_n b_{n+2} =: D \ge 0$ at any root. The auxiliary lemma `im_zero_of_sq_eq_nonneg_real` ($w^2 = D \in \mathbb{R}_{\ge 0} \Rightarrow \operatorname{Im} w = 0$) forces $b_{n+2}z + b_{n+1} \in \mathbb{R}$, hence (as $b_{n+2} \neq 0$ real) $\operatorname{Im} z = 0$. No square roots and no zero-locus input are used. □

Nondegeneracy $b_{n+2} \neq 0$ is supplied by (5) ($M_n > 0$). The analytic input is classical. Csordas, Norfolk and Varga [7] proved Pólya's conjectured Turán inequalities for the ξ coefficients. Dimitrov and Lucas [14] proved the higher-order (degree-3) Turán inequalities without assuming RH. Theorem 8 is the formal bridge for the diagnostic normalization, not a new inequality or a proof of the classical Jensen criterion. For that criterion the correct quadratic condition is

$$\gamma_{n+1}^2 \ge \gamma_n \gamma_{n+2} \quad \Longleftrightarrow \quad \mu_{n+1}^2 \ge \frac{n+2}{n+1} \mu_n \mu_{n+2}. \tag{11}$$

It is stronger than plain log-concavity of μ. The classical results on degrees two and three concern the factorial-normalized polynomials $J_{d,n}^{\mathrm{cl}}$ [15]; the displayed Lean diagnostic theorem does not identify those two families.

4.5 An independent operator-positivity route

The naive Hilbert–Pólya operator must be self-adjoint for the ambient inner product—the obstruction. The pseudo-Hermitian viewpoint [10, 11] keeps only a *positive metric*.

Definition 9 (`PseudoHermitianForm`)**.** For a $\mathbb{C}$-linear T on a $\mathbb{C}$-vector space E, the formal interface is a map $\eta\colon E \times E \to \mathbb{C}$ with $\eta(cx, y) = c\eta(x, y)$, $\eta(x, cy) = \bar{c}\eta(x, y)$, $\eta(Tx, y) = \eta(x, Ty)$, and $\eta(x, x) \neq 0$ for every $x \neq 0$. A positive-definite sesquilinear form supplies these properties. The Lean structure records only these scalar laws, invariance, and diagonal nonvanishing; it does not require additivity or Hermitian symmetry. Diagonal nonvanishing is stronger than the usual nondegeneracy of a form.

Theorem 10 (`PseudoHermitianForm.eigenvalue_conj_eq` and `riemannHypothesis_of_pseudoHermitianWitness`; verified)**.** *An η-self-adjoint operator has real eigenvalues. Consequently, if the every critical-strip zero s has a nonzero eigenvector with eigenvalue $\lambda(s) = -i\,(s - \frac{1}{2})$ (real iff $\operatorname{Re} s = \frac{1}{2}$), then RH holds.*

Proof (mechanized). For an eigenpair $Tv = \lambda v$, $v \neq 0$: $\lambda\,\eta(v, v) = \eta(Tv, v) = \eta(v, Tv) = \bar{\lambda}\,\eta(v, v)$, and $\eta(v, v) \neq 0$, so $\lambda = \bar{\lambda}$. The proof uses only the form laws—no inner product, completeness or boundedness. It is a separate conditional implication; logical independence from other RH criteria is not asserted. The witness requires a zero-to-eigenvector map, not equality with the full operator spectrum. No construction of such a witness is supplied. Closedness, domains and continuous spectrum would be additional obligations for an unbounded Hilbert-space realization. □

5 Numerical consistency checks

Coefficient enclosures. The coefficients $\mu_0, \ldots, \mu_{120}$ are enclosed by rigorous balls computed with FLINT/Arb [24] (python-flint 0.8.0, 4096 working bits). The computation uses rigorous special-function Taylor-series algorithms; the independent theta-integral check uses explicit analytic tail bounds. Every enclosure has relative accuracy of at least 2317 bits and exact rational endpoints. The first nine coefficients agree with an independent theta-integral enclosure. Data and generator are in `research/certified_xi/coefficients_120.json` (SHA-256 `adf3418d...63b11`) of the archived release. The June draft instead used two floating-point Cauchy-contour computations (620 and 520 digits), which agree to ≥ 236 digits. That is a consistency check, not an enclosure, and is superseded here.

Certified claims. The script `paper/certify_reduction_minors.py` works at 8192 bits. It certifies in ball arithmetic, so that each output ball lies strictly in $(0, \infty)$:

(i) $\mu_n > 0$ for $0 \leq n \leq 120$;

(ii) strict log-concavity $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ for $0 \leq n \leq 118$;

(iii) every contiguous zero-extended Toeplitz minor $\det[\mu_{m+i-j}]_{i,j<k}$ with $1 \leq k \leq 16$ and $m + k - 1 \leq 120$ is strictly positive. These are 1816 minors, and the largest relative radius of a determinant ball is 6.5×10^{-669}, at $k = 16$, $m = 105$.

The certificate output, including the input hash, is stored in `paper/rh_reduction_minor_certificate.json`. The trust boundary is FLINT/Arb together with the classical identities behind the enclosures. These are not Lean certificates. The June draft also plotted floating-point roots for $d \leq 6$ and $n \leq 114$. The source script uses $Q_{d,n}$, not $J_{d,n}^{\mathrm{cl}}$; that uncertified diagnostic plot is omitted from this revision and is not used as evidence for the classical Jensen criterion.

What the checks do and do not show. Nonnegativity in (iii) is already a theorem: Katkova's finite-order result covers every minor of order ≤ 43, at every index [13]. Claim (ii) is a special case of the infinite log-concavity of (μ_n) proved in [20]. That proof is a preprint and is computer-assisted. For Jensen polynomials, hyperbolicity is known for $d \leq 3$ at every shift [7, 14], and for every fixed d at all sufficiently large shifts [8, 15]. The computations above are therefore calibration of conventions and margins: strictness with explicit error control in a finite range. They are not discoveries of previously unknown positivity. The figures in this revision are regenerated directly from the certified coefficient artifact using the corrected normalization dictionary.

The classical Jensen baseline extends far beyond degree three: Corollary 1.3 of [15], using a finite verification of low zeros, gives hyperbolicity for all shifts when $d \leq 9.36 \times 10^{20}$. This is a fixed-degree result for $J^{\mathrm{cl}}_{d,n}$, not an all-degrees theorem and not a result about the unnormalized diagnostic $Q_{d,n}$.

Figures 3–5 use midpoints of the certified coefficient balls. The determinant signs are checked again before plotting; the raster figures display the data and are not themselves certificates. The figure generator is `paper/generate_reduction_figures.py`.

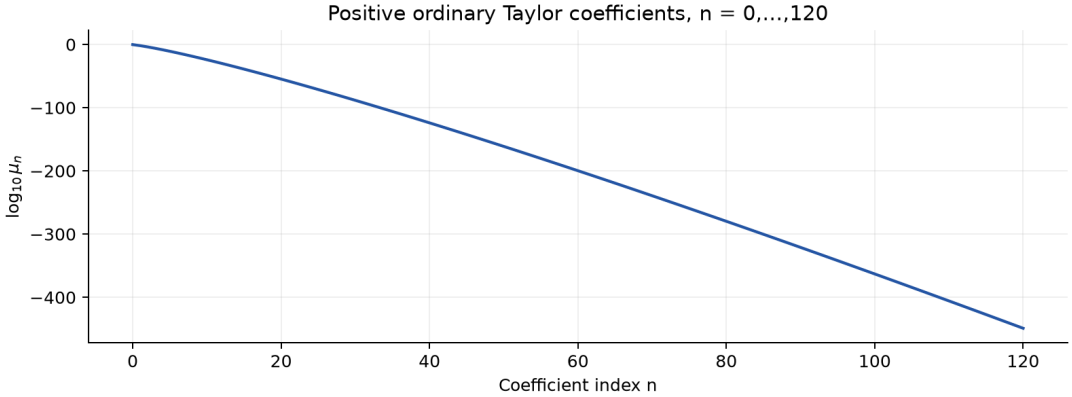


Figure 3: Ordinary sign-normalized coefficients μ_n, $0 \leq n \leq 120$, plotted as $\log_{10} \mu_n$ to avoid floating-point underflow. All input balls are strictly positive. The plotted midpoint does not display the much smaller enclosure radius.

Interpretation. These checks validate the conventions and finite strictness margins, within classical finite-order positivity ranges. They cover contiguous blocks only, not every arbitrary minor involving the first 121 coefficients. The unresolved issue is an all-orders statement, with control also outside the large-shift regions of external asymptotic results.

6 The open frontier

The reductions convert "zeros on a line" into positivity statements about the explicit positive kernel Φ and an explicit operator metric—objects amenable to tools foreign to zero-counting.

A. Uniform total positivity. Prove (8) for all orders and arbitrary increasing row and column selections. This is RH-equivalent, given the classical inputs of Theorem 5. Proving strict contiguous positivity for every k, m, followed by an applicable closure theorem, is one possible

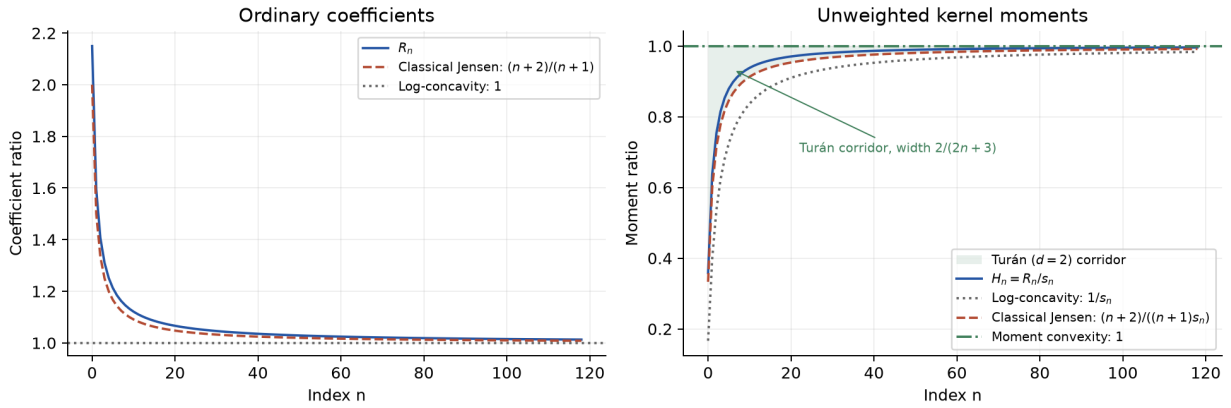


Figure 4: Two distinct normalizations of the order-two inequality for $0 \le n \le 118$. Left: the coefficient ratio R_n compared with 1 and the stronger classical Jensen threshold $(n+2)/(n+1)$. Right: the moment ratio $H_n = R_n/s_n$ compared with its equivalent thresholds $1/s_n$ and $(n+2)/((n+1)s_n)$, and the moment-convexity upper bound 1. Since $(n+2)/((n+1)s_n) = (2n+1)/(2n+3)$ exactly, the shaded band $(2n+1)/(2n+3) \le H_n \le 1$ is the order-two Jensen (Turán) condition. Its width is $2/(2n+3)$. It is a theorem [7], not an RH-strength statement. Curves use coefficient-ball midpoints; only the plain log-concavity claim is part of the reported minor certificate.

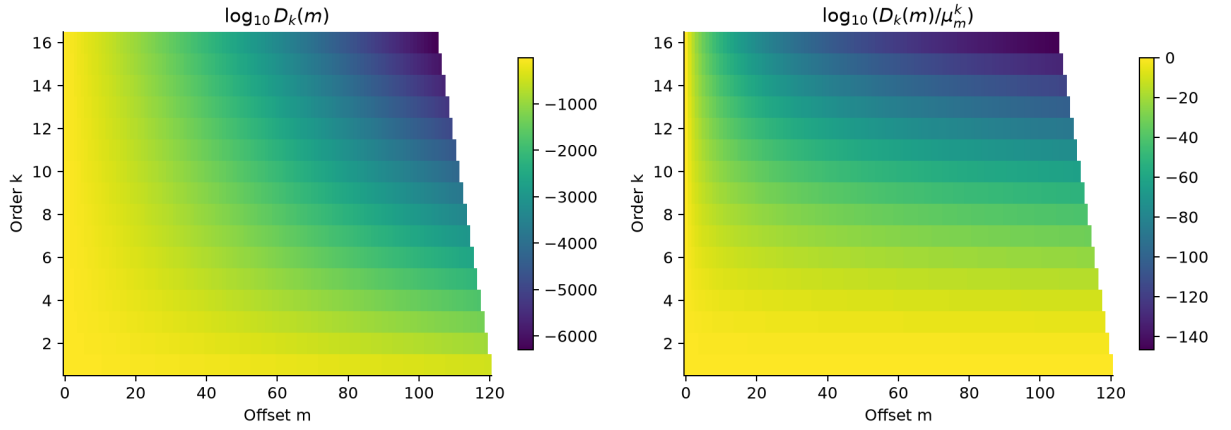


Figure 5: Certified contiguous Toeplitz determinants $D_k(m) = \det[\mu_{m+i-j}]_{i,j<k}$ for $1 \le k \le 16$ and $0 \le m \le 121-k$, with zero extension at negative indices. Left: $\log_{10} D_k(m)$ for all 1816 tested blocks. Right: $\log_{10}(D_k(m)/\mu_m^k)$ on the same domain. Blank cells require coefficients beyond index 120 and are not tested. Every included determinant ball lies strictly in $(0, \infty)$.

sufficient route. Every *fixed* order up to 43 is classical [13]. A recent preprint claims strict positivity of every contiguous minor of order $r \geq 2$ at all shifts $m \geq 10^{18} r^3$, using certified saddle-point estimates [18]. The open content is therefore the complementary region of unbounded order and moderate shift. In that preprint's convention $a_n = \mu_n/(8 \cdot 4^n)$, so its transposed contiguous determinant is $8^{-r} 4^{-mr} D_r(m)$; the positive scaling preserves signs. This normalization transfer does not verify the preprint's analytic estimates. The continuous de Bruijn–Newman kernel is not PF_5 [19], so a continuous-kernel total-positivity shortcut is unavailable. Classical tools are Karlin total positivity [6], Schoenberg Pólya-frequency functions, and variation-diminishing criteria for the representation $F(z) = 2\int_0^\infty \Phi(u)\cosh(u\sqrt{z})\,du$.

B. Quantitative margins. Obtain effective bounds on $R_n - 1$ and the corresponding moment margin $H_n - 1/s_n = (R_n - 1)/s_n$. The elementary slack gap is of order $1/n$, as in (10). Such fixed-order estimates alone cannot establish full PF. Tools: the Griffin–Ono–Rolen–Zagier Hermite asymptotics [8] and their effective form [15]. Farmer argues that Jensen-polynomial asymptotics alone are not a plausible route to RH [16].

C. μ log-concavity (settled). The June draft listed $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ as open. By (5) it is equivalent to $M_{n+1}^2/(M_n M_{n+2}) > 1/s_n = \frac{(2n+1)(2n+2)}{(2n+3)(2n+4)}$. The Turán inequalities of Csordas–Norfolk–Varga [7] give $M_{n+1}^2 > \frac{2n+1}{2n+3} M_n M_{n+2}$, which is stronger. A recent preprint proves much more, strict *infinite* log-concavity of (μ_n) [20]. Like Jensen asymptotics, this is a consequence of RH and does not imply it.

D. Operator witness. Construct (E, T, η) and a nonzero eigenvector for every critical-strip zero with eigenvalue $-i(s - 1/2)$; a formal operator expression alone does not supply this witness (cf. [11]).

The de Bruijn–Newman programme provides a dynamical handle on the same object. Writing its constant as Λ_{DN} to distinguish it from completed zeta, the established bounds are $\Lambda_{\mathrm{DN}} \geq 0$ [9] and $\Lambda_{\mathrm{DN}} \leq 0.2$ [17], while RH $\iff$ $\Lambda_{\mathrm{DN}} \leq 0$.

Remark 11 (A conditional finite-order improvement). Katkova's order 43 uses two inputs. The first is that nontrivial zeros satisfy $|\operatorname{Re}\rho - \frac{1}{2}| \leq \frac{1}{2}$. The second is that no zero has $|\operatorname{Im}\rho| < \gamma_1 = 14.1347\ldots$. In the variable $z = (s - \frac{1}{2})^2$, a zero $\frac{1}{2} + x + i\gamma$ lies at angle $2\arctan(|x|/\gamma)$ from the negative axis. Schoenberg's sector theorem yields PF_m when this angle is at most $\pi/(m+1)$. A recent manuscript claims that $\zeta(s) \neq 0$ for $\operatorname{Re} s > 7/8$ [21]. The external collection advertises a Lean formalization, but neither that formalization nor its analytic proof is audited here. It is treated only as an external claim, not as a verified input to this manuscript. If the claim holds, then $|x| \leq \frac{3}{8}$, the maximal angle drops from 0.07072 to 0.05305, and the same argument gives PF_{58} in place of PF_{43}. This is a constant improvement only. Verification of RH to a specified finite height can improve the resulting finite-order bound; a single finite verification does not establish all orders. The remark is not used elsewhere in this paper.

6.1 A Davenport–Heilbronn sector control

To test whether the sufficient sector rank predicts the failure rank of the full coefficient sequence, use the conductor-five Davenport–Heilbronn function [23], with its entire odd completion

$$\Lambda_H(s) = (5/\pi)^{s/2}\Gamma((s+1)/2)H(s), \qquad F_H(w) = \Lambda_H(1/2 + \sqrt{w})/\Lambda_H(1/2).$$

The ordinary coefficients of F_H, rather than their factorial multiples, are used below. These are new computer-assisted results in the working tree, separate from the pinned Lean results described in the next section.

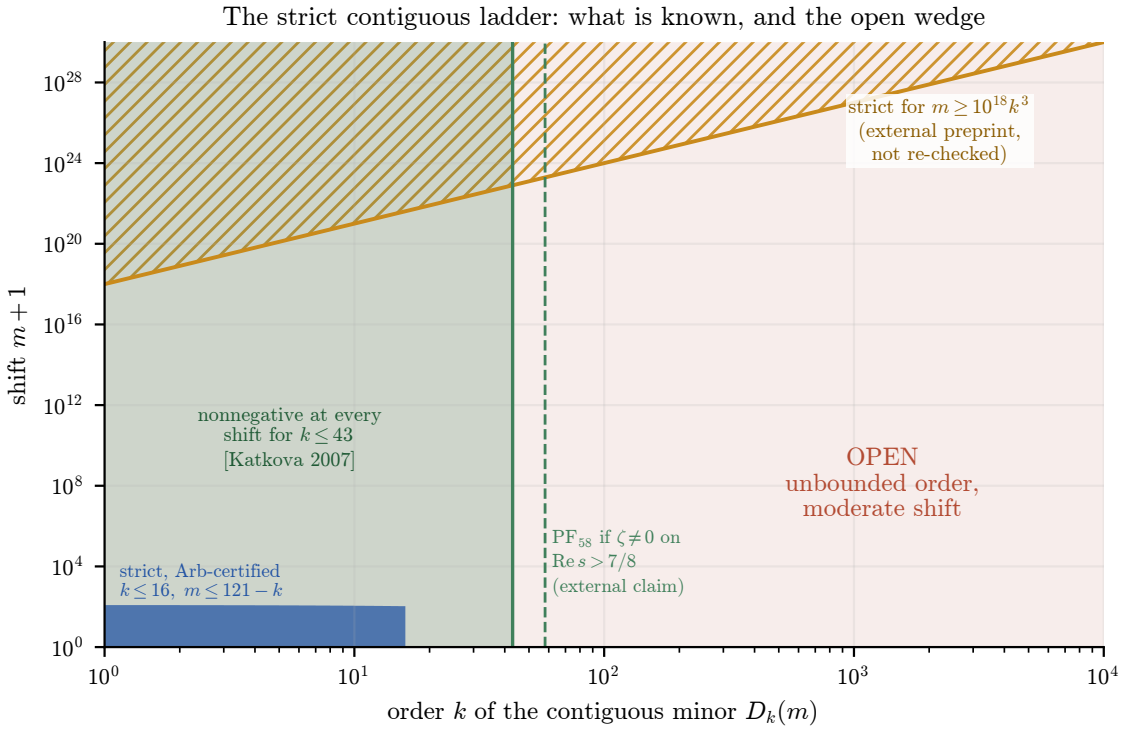


Figure 6: The order–shift plane of the contiguous determinants $D_k(m)$, on logarithmic axes. Blue: the strict positivity certified in §5 ($k \leq 16$, $m \leq 121 - k$). Green: every minor of order $k \leq 43$ is nonnegative at every shift [13]; this is not strictness. Hatched: the external cubic-wedge claim $m \geq 10^{18}k^3$ [18], not re-checked here. The dashed line is the conditional PF_{58} of Remark 11. The rose region, unbounded order at moderate shift, is where frontier A lives.

An absolute Dirichlet-tail bound below 0.737591 at $\operatorname{Re} s = 3/2$ and the completed functional equation give $|\operatorname{Re}\rho - 1/2| < 1$. An Arb argument-principle computation counts 249 zeros in $[-1/2, 3/2] + i(0, 340)$; 235 critical-line sign brackets and seven reflected pairs of simple Rouché disks account for the entire count. Comparing all seven right-hand disks and the higher-zero bound $2\arctan(1/340)$ proves that $0.80851718\ldots + 85.69934849\ldots i$ has the steepest folded angle ϕ, with

$$436 < \pi/\phi = 436.3348656\ldots < 437.$$

The genus-zero product and Schoenberg's sufficient sector theorem therefore give PF_{435}. For the isolated quadratic factor, the shift-one rank-k minor is $U_k(\cos\phi) = \sin((k+1)\phi)/\sin\phi$, first negative at $k = 436$.

Nevertheless every one of the 18 291 full-function solid minors with $430 \le r \le 520$ and $0 \le m \le 200$ is strictly positive at 32 768 bits, with no unresolved signs. The complete condensation sign stream is replayed, and 24 selected high-rank minors are checked independently using complementary determinants. Thus the isolated factor's sharp rank does not predict a failure on this full-function grid. Neither the globally first negative minor nor PF_{520} follows from a bounded solid-minor scan. Exact balls, analytic dependencies, source hashes and reproduction instructions are in `research/prime_spectral/RESEARCH_ROUND_2026_10_12.md`.

7 Methodology, reproducibility, and limitations

Verification. Every Lean theorem and named input cited in this paper exists at commit `c749786972306bfe6be8890d78972bd9a2325f0c` of https://github.com/sciencenerd-des/reinmann. The related two-routes manuscript is identified by DOI 10.5281/zenodo.23187646 [25]; the source pin for the present formal claims is the Git commit, not a claim that this revised working manuscript has been published. The pinned toolchain is Lean `v4.30.0`; the exact Mathlib revision and Lake manifest hash are given in Appendix A. Only the modules and statements listed in Table 2 are needed for the formal results presented here. The repository scripts `scripts/safe-verify.sh` and `scripts/verify_axiom_clean.sh` reject `sorry`, `admit`, added axioms and similar shortcuts. For this revision, on 10 October 2026, the focused four-module build at the source pin completed successfully. The shortcut scan passed for its 77 project Lean files, and the focused audit printed the types and axiom dependencies of 18 manuscript theorems; each had exactly the three foundational axioms listed above. The finite minor replay also reproduced all 1816 positive determinants. These checks do not assert a fresh full build of the later working tree or replay the external preprint proofs.

Named external inputs (never asserted). `PolyaJensenBridge`, `XiToeplitzTPToScaledFiniteBridge`, `XiLaguerrePolyaClosureBridge`, `XiMomentKernelRep`, `XiToeplitzTPFromRH`. Each is a standard classical theorem; formalizing them in Mathlib is future work.

Theorem ledger. Table 2 distinguishes kernel-checked implications, explicit unproved hypotheses, numerical certificates, and open targets. In particular, `XiMomentKernelRep` records existence of positive numbers satisfying the coefficient identities; its Lean definition does not construct the function Φ or its integral. Those analytic identities are classical inputs to the numerical and mathematical interpretation.

Table 2: Result and dependency ledger for this manuscript. "Checked" describes the stated implication, including all its hypotheses.

Result	Source module	Status and required inputs
Critical-line zero identity	`JensenProgram.lean`	Checked; `Xi_eq_zero_iff_riemannZeta`, no PF hypothesis.
PF low-order extraction	`XiToeplitzPositivity.lean`	Checked algebra under `XiToeplitzTotalPositive`; it does not prove that hypothesis.
PF/RH tightness	`RHReductionCapstone.lean`	Checked under the three forward bridges and `XiToeplitzTPFromRH`.
Moment determinant scaling	`XiMomentKernel.lean`	Checked for contiguous minors under the coefficient representation h_M; kernel construction is external.
Diagnostic quadratic	`JensenTuranBridge.lean`	Checked under `XiTuran2` and $b_{n+2} \neq 0$; concerns $Q_{2,n}$, not $J^{\mathrm{cl}}_{2,n}$.
Eigenvalue/RH implication	`PseudoHermitian.lean`	Checked under the form laws and zero-indexed nonzero eigenvector witness; no operator is constructed.
ASW/Edrei and closure	`XiToeplitzTPTo ScaledFiniteBridge`, `XiLaguerrePolyaClosureBridge`	Named external classical inputs; no proof of these propositions is supplied by the capstone.
Pólya–Jensen and converse	`PolyaJensenBridge`, `XiToeplitzTPFromRH`	Named external classical inputs, with the factorial-normalized Jensen convention.
Positive moment representation	`XiMomentKernelRep`	Named external input; its existential moment sequence is not a formalized quadrature.
Finite minor positivity	`rh_reduction_minor_ certificate.json`	Arb-certified, under exact coefficient enclosures and analytic formulas; not Lean-checked.
Recent positivity results	References [18, 19, 20]	External preprint results; neither their proofs nor ancillary certificates are replayed here.
All-orders PF / operator witness	Equation (8) / Theorem 10	Open targets; finite tests and named bridges do not establish them.

Limitations.

1. No proof of RH: the PF capstone still requires the all-orders positivity input, as well as its named classical bridges. The operator reduction requires the zero-indexed eigenvector witness. No combination of the classical background theorems alone supplies these inputs.

2. The numerics are finite. Ball arithmetic certifies positivity only for $n \leq 120$ and Toeplitz order $k \leq 16$, under the FLINT/Arb trust boundary. The legacy diagnostic root observations for $d \leq 6$ are floating-point only and are omitted. None of this can establish the uniform $(k, n \to \infty)$ statement that is RH-equivalent, and the certified range lies inside what known theorems already cover.

3. The pseudo-Hermitian existence claim is entirely open and unevidenced.

4. Equivalence-chain reformulations (e.g. "≤ 1 zero per height") are correct but RH-equivalent, hence reorganization rather than progress.

8 Related work

Classical background. Classical sources are Pólya [2, 3], Aissen–Schoenberg–Whitney [4], Edrei [5] and Karlin [6]. For the ξ coefficients specifically, Katkova [13] gives the Pólya-frequency formulation and finite-order positivity. The Turán and higher-order Turán inequalities are due to Csordas–Norfolk–Varga [7] and Dimitrov–Lucas [14].

Jensen polynomials and heat flow. Jensen-polynomial asymptotics are developed in [8, 15], with the critique of [16]. On the de Bruijn–Newman constant see [9, 17].

Recent work on the coefficients and kernel. Three 2026 preprints concern these objects: a tail wedge for contiguous Toeplitz minors [18], certified failure of PF_5 for the continuous kernel [19], and infinite log-concavity [20]. A claimed zero-free half-plane $\operatorname{Re} s > 7/8$ [21] bears on finite-order constants only (Remark 11).

Operators and formalization. For the pseudo-Hermitian viewpoint see Mostafazadeh [10] and Bender–Brody–Müller [11]. The formal foundations are Mathlib [12] and its zeta and L-function library [22]. Spectral and adelic perspectives (Berry–Keating, Connes) and the Weil and Li positivity criteria are classical antecedents of the operator and positivity viewpoints used here.

9 Conclusion

The checked contribution is a conditional deduction architecture: full PF positivity and named analytic bridges imply RH, and a named converse makes that reduction tight. The accompanying coefficient identities distinguish ordinary Taylor coefficients, factorial-normalized Jensen coefficients and kernel moments. The operator theorem supplies a separate eigenvector-based implication. The finite Arb certificates validate conventions and strictness in the stated bounded domain.

Further work has three distinct obligations: formalize the classical analytic bridges with their complete hypotheses; establish all-orders positivity rather than a fixed-order or large-shift fragment; and obtain independent review of the analytic identification and numerical enclosure formulas. None is discharged by the conditional theorem or finite data.

Disclosure of AI-assisted tools

Large language model assistants from OpenAI and Anthropic were used for exposition, literature triage, numerical code scaffolding and consistency checks. This revision also used Codex for manuscript editing, replay scripts and figure regeneration. These tools are not authors. Formal correctness claims rest on the stated Lean theorem types and kernel checks; numerical claims rest on the specified enclosure algorithms and Arb trust boundary. The author remains responsible for reviewing the final text, mathematical interpretation, references and declarations before submission.

Acknowledgements

The author thanks the Lean and Mathlib communities for the formal-mathematics infrastructure on which this development depends.

A Reproducibility specification

Formal source pin. The source commit is `c749786972306bfe6be8890d78972bd9a2325f0c`, the toolchain is `leanprover/lean4:v4.30.0`, and the Mathlib commit is `c5ea00351c28e24afc9f0f84379aa41082b1188f`. The SHA-256 of `lake-manifest.json` at that commit is `a68511672a9f3196d87b941919c1a9b53a2347723ba6afe4ceaef326bbb0a961`. Verification should use a separate checkout at this commit rather than silently include later, uncommitted research modules. From that checkout:

```
lake exe cache get
lake build Reinmann.RHReductionCapstone Reinmann.JensenTuranBridge \
  Reinmann.XiMomentKernel Reinmann.PseudoHermitian
bash scripts/safe-verify.sh Reinmann/RHReductionCapstone.lean \
  Reinmann/JensenTuranBridge.lean Reinmann/XiMomentKernel.lean \
  Reinmann/PseudoHermitian.lean
bash scripts/verify_axiom_clean.sh
```

The last command audits the broader archived development. The focused audit file supplied with this revision, `paper/reduction_axiom_audit.lean`, prints both theorem types and axiom dependencies for this paper; copy it into the pinned checkout and run `lakeenvleanpaper/reduction_axiom_audit.lean`. The expected dependency sets are subsets of {propext, Classical.choice, Quot.sound}, with no `sorryAx` or custom axioms. The theorem types must still be inspected for explicit assumptions. Successful source scanning alone is insufficient.

Numerical artifacts. The input is `research/certified_xi/coefficients_120.json`, with SHA-256 `adf3418dbf0e7b823f204e527155d240f52034a8027fb0ecb058007a27963b11`. The generator `experiments/certified_xi/coefficients.py` uses Python 3.12, `python-flint==0.8.0`, 4096 working bits, and an independent theta check through $n = 8$. The minor checker uses 8192 bits. From the repository root the existing artifact can be replayed without replacing it:

```
uv run --with python-flint==0.8.0 --python 3.12 \
  python paper/certify_reduction_minors.py > /tmp/reduction-replay.json
uv run --with python-flint==0.8.0 --with matplotlib --python 3.12 \
  python paper/generate_reduction_figures.py
```

Expected output is `all_certified:true`, 121 positive coefficient balls, 119 strict log-concavity inequalities, 1816 positive determinant balls, and no uncertified blocks. The number of determinants is

$$\sum_{k=1}^{16}(122-k) = 1816. \tag{12}$$

The coefficient generator can be rerun to a separate output path with `--max-n120--bits4096--accuracy128--theta-max-n8`; its complete invocation and enclosure-tail derivation are in `experiments/certified_xi/README.md`. Recomputed ball endpoints may differ while enclosing the same values; hash equality identifies the published input artifact, not every valid regeneration.

Document build and revision evidence. Run `tectonicrh_reduction_paper.tex` from `paper/`; a conventional pdfLaTeX build also requires repeated passes for references and the contents. Figures and the inline bibliography are required alongside the source. The revision's validation

record is `paper/rh_reduction_revision_validation.json`; it records actual command outcomes and their scope. The publication preparation checklist is `paper/RH_REDUCTION_READINESS.md`. The revision metadata in `paper/zenodo_rh_reduction.json` is local preparation, not evidence of a remote upload or publication.

References

[1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin Akad., 1859.

[2] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. **48** (1926), 305–317.

[3] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Sel. Math.-Fys. Medd. **7** (1927).

[4] M. Aissen, I. J. Schoenberg, A. Whitney, *On the generating functions of totally positive sequences I*, J. Anal. Math. **2** (1952), 93–103.

[5] A. Edrei, *On the generating functions of totally positive sequences II*, J. Anal. Math. **2** (1953), 104–109.

[6] S. Karlin, *Total Positivity, Vol. I*, Stanford Univ. Press, 1968.

[7] G. Csordas, T. S. Norfolk, R. S. Varga, *The Riemann Hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541.

[8] M. Griffin, K. Ono, L. Rolen, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), 11103–11110.

[9] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6.

[10] A. Mostafazadeh, *Pseudo-Hermiticity versus PT-symmetry*, J. Math. Phys. **43** (2002), 205–214.

[11] C. M. Bender, D. C. Brody, M. P. Müller, *Hamiltonian for the zeros of the Riemann zeta function*, Phys. Rev. Lett. **118** (2017), 130201.

[12] The mathlib Community, *The Lean mathematical library*, Proc. CPP 2020, 367–381.

[13] O. M. Katkova, *Multiple positivity and the Riemann zeta-function*, Comput. Methods Funct. Theory **7** (2007), 13–31; arXiv:math/0505174.

[14] D. K. Dimitrov, F. R. Lucas, *Higher order Turán inequalities for the Riemann ξ-function*, Proc. Amer. Math. Soc. **139** (2011), 1013–1022.

[15] M. Griffin, K. Ono, L. Rolen, J. Thorner, Z. Tripp, I. Wagner, *Jensen polynomials for the Riemann xi function*, arXiv:1910.01227.

[16] D. W. Farmer, *Jensen polynomials are not a plausible route to proving the Riemann Hypothesis*, Adv. Math. **411** (2022), 108781.

[17] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Res. Math. Sci. **6** (2019), Paper No. 31.

[18] W. Michałowski, *An explicit uniform cubic wedge for consecutive Toeplitz minors of the Riemann xi coefficients*, preprint (2026), arXiv:2607.16795.

[19] W. Michałowski, *On the Pólya frequency order of the de Bruijn–Newman kernel: certified failure at order five*, preprint (2026), arXiv:2602.20313.

[20] Y. Liu, J. Mao, *Infinite log-concavity of the Taylor coefficients of the Riemann xi-function*, preprint (2026), arXiv:2610.04972.

[21] OpenAI, *The quasi-Riemann hypothesis: a zero-free half-plane* $\operatorname{Re} s > 7/8$, manuscript dated 30 September 2026, with Lean formalization, `https://github.com/openai/math/blob/main/lean/docs/003.md` (scope documentation accessed 10 October 2026).

[22] D. Loeffler, M. Stoll, *Formalizing zeta and L-functions in Lean*, Annals of Formalized Mathematics **1** (2025).

[23] E. Bombieri, A. Ghosh, *Around the Davenport–Heilbronn function*, Russian Math. Surveys **66** (2011), 221–270, doi:10.1070/RM2011v066n02ABEH004740.

[24] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.

[25] B. Mondal, *Two machine-checked conditional routes to the Riemann Hypothesis*, version 0.2.0, Zenodo (2026), doi:10.5281/zenodo.23187646.

[26] B. Mondal, *Infinite Fourier tails and high-mode coercivity for the semilocal Weil quadratic form*, version 1.0.0, Zenodo (2026), doi:10.5281/zenodo.23187645.